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# Comptia N+ Assignment

## [N+ Network Fundamental and building networks]

### Hands-on Subnetting

**Divide a network into subnets and assign IP addresses to devices.Task: Document the steps to create a subnet plan.**

Task:

❖ **1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.**

➢ **Aim**

- To divide the given network 192.168.10.0/24 into four equal subnets and determine the subnet mask, network address, and broadcast address of each subnet.

➢ **Introduction**

- Subnetting is a networking technique used to divide a large network into smaller logical networks called subnets. It helps in efficient utilization of IP addresses, reduces unnecessary broadcast traffic, and improves network management and organization.
- In this assignment, the network 192.168.10.0/24 is divided into four equal subnets.

- **Given Information**
  - **Network Address:** 192.168.10.0/24
  - **Required Subnets:** 4
  - **Original CIDR:** /24
  - **Original Subnet Mask:** 255.255.255.0

- **Subnetting Calculation**
- **Step 1: Calculate the borrowed bits**
  - ✓ The formula for calculating the number of subnets is:
  - ✓ Number of Subnets = (2^n)
  - ✓ Where (n) is the number of borrowed host bits.
  - ✓ We need 4 subnets:
  - ✓ (2^n = 4)
  - ✓ (2^2 = 4)
  - ✓ Therefore, **2 bits are borrowed** from the host portion.
- **Step 2: Calculate the new CIDR notation**
  - ✓ Original CIDR = /24
  - ✓ Borrowed bits = 2
  - ✓ New CIDR = 24 + 2 = /26
  - ✓ Therefore, the new subnet prefix is **/26**.
- **Step 3: Calculate the new subnet mask**
  - ✓ The subnet mask for /26 is:
  - ✓ **255.255.255.192**
- **Step 4: Calculate the number of addresses per subnet**
  - ✓ Remaining host bits:
  - ✓ (32 - 26 = 6)
  - ✓ Total IP addresses per subnet:

- ✓ (2^6 = 64)
- ✓ Usable host addresses per subnet:
- ✓ (64 - 2 = 62)
- ✓ Therefore, each subnet contains **64 total IP addresses and 62 usable host addresses**.

➢ **Step 5: Calculate the block size**

- ✓ Block size = 256 − 192
- ✓ Block size = **64**
- ✓ Therefore, the subnet network addresses increase by 64 in the last octet.
- ✓ The four network addresses are:
  - 192.168.10.0
  - 192.168.10.64
  - 192.168.10.128
  - 192.168.10.192

➢ **Subnet Plan**

| Subnet | Network Address | Subnet Mask | Broadcast Address |
|---|---|---|---|
| 1 | 192.168.10.0/26 | 255.255.255.192 | 192.168.10.63 |
| 2 | 192.168.10.64/26 | 255.255.255.192 | 192.168.10.127 |
| 3 | 192.168.10.128/26 | 255.255.255.192 | 192.168.10.191 |
| 4 | 192.168.10.192/26 | 255.255.255.192 | 192.168.10.255 |

- **Valid Host Address Ranges**

| Subnet | First Usable IP | Last Usable IP | Usable Hosts |
|---|---|---|---|
| 1 | 192.168.10.1 | 192.168.10.62 | 62 |
| 2 | 192.168.10.65 | 192.168.10.126 | 62 |
| 3 | 192.168.10.129 | 192.168.10.190 | 62 |
| 4 | 192.168.10.193 | 192.168.10.254 | 62 |

- The network address identifies each subnet, while the broadcast address is used to communicate with all devices in that subnet. These two addresses cannot be assigned to ordinary host devices.

- **Steps to Create a Subnet Plan**
  1. Identify the original network address and CIDR prefix.
  2. Determine the required number of subnets.
  3. Calculate the number of bits to borrow using the formula (2^n).
  4. Add the borrowed bits to the original CIDR prefix.
  5. Determine the new subnet mask and block size.
  6. Calculate the network and broadcast addresses for each subnet.
  7. Identify the valid host address range in every subnet.
  8. Assign unique IP addresses from the usable range to devices within each subnet.

- **Example of IP Address Assignment**
- The following example shows how IP addresses can be assigned to devices in the first subnet.

| Device | Assigned IP Address | Subnet Mask |
|---|---|---|
| Router/Gateway | 192.168.10.1 | 255.255.255.192 |
| Laptop | 192.168.10.2 | 255.255.255.192 |
| Desktop Computer | 192.168.10.3 | 255.255.255.192 |
| Printer | 192.168.10.4 | 255.255.255.192 |

- These addresses are examples of assignments within the first subnet. The default gateway is assumed to be 192.168.10.1.

- **Result**
- The network 192.168.10.0/24 was successfully divided into four equal subnets using the /26 CIDR prefix and subnet mask 255.255.255.192. Each subnet contains 64 total IP addresses and 62 usable host addresses.

❖ **2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Make sure not to use the network or broadcast address for any device.**

➢ **Aim**

- To assign valid and unique IP addresses to five network devices within a subnet while avoiding the network and broadcast addresses.

➢ **Introduction**

- IP addressing is an important part of computer networking. Each device connected to a network must have a unique IP address to communicate with other devices. When assigning IP addresses, the network address and broadcast address must not be assigned to ordinary host devices.
- In this assignment, five devices are assigned valid IP addresses from the first subnet created in the previous subnetting exercise.

➢ **Given Network Details**

- Original Network: 192.168.10.0/24
- Selected Subnet: 192.168.10.0/26
- Subnet Mask: 255.255.255.192
- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 to 192.168.10.62
- Total Usable Host Addresses: 62

- **IP Address Assignment Table**

| No. | Device | Assigned IP Address | Subnet Mask |
|---|---|---|---|
| 1 | Router/Gateway | 192.168.10.1 | 255.255.255.192 |
| 2 | Laptop 1 | 192.168.10.2 | 255.255.255.192 |
| 3 | Laptop 2 | 192.168.10.3 | 255.255.255.192 |
| 4 | Printer | 192.168.10.4 | 255.255.255.192 |
| 5 | Mobile Phone | 192.168.10.5 | 255.255.255.192 |

- **Explanation**
  1. Router/Gateway: The IP address 192.168.10.1 is assigned to the router, which acts as the gateway for devices in this subnet.
  2. Laptop 1: The IP address 192.168.10.2 is assigned to the first laptop.
  3. Laptop 2: The IP address 192.168.10.3 is assigned to the second laptop.
  4. Printer: The IP address 192.168.10.4 is assigned to the printer so that devices can communicate with it over the network.
  5. Mobile Phone: The IP address 192.168.10.5 is assigned to the mobile phone.
- All five IP addresses are unique and belong to the valid host range of the selected subnet.

- **Verification**
  - The network address 192.168.10.0 has not been assigned to any device.
  - The broadcast address 192.168.10.63 has not been assigned to any device.
  - All assigned IP addresses fall within the valid host range of 192.168.10.1 to 192.168.10.62.
  - No two devices share the same IP address.
  - All devices use the same subnet mask: 255.255.255.192.

- **Result**
- Five network devices were successfully assigned unique and valid IP addresses within the subnet 192.168.10.0/26. None of the devices uses the network address or broadcast address.

❖ **3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.**

➢ **Aim**

- To draw a subnet plan showing four subnets, their valid IP address ranges, and the devices connected to each subnet.

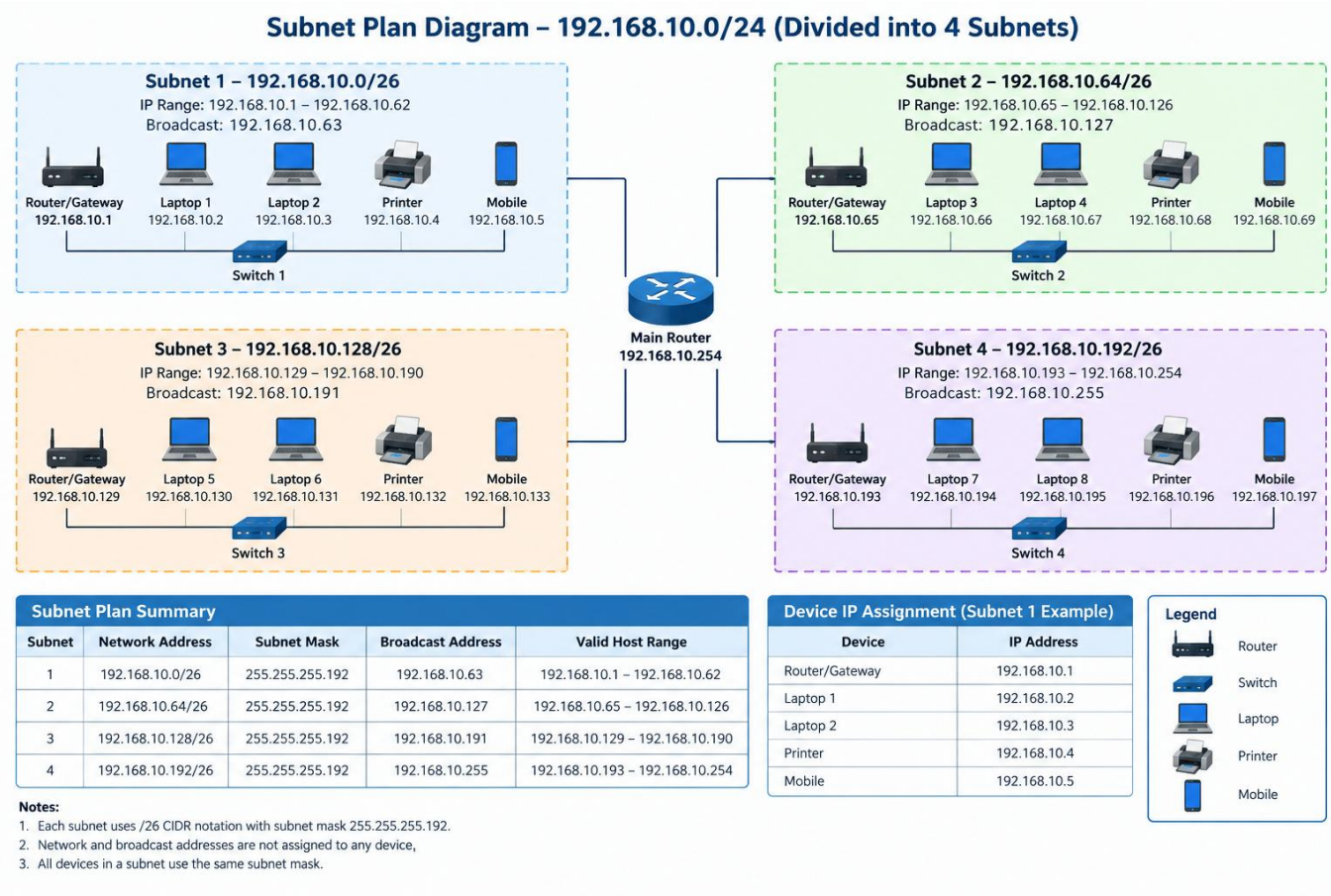
➢ **Introduction**

- Subnetting divides a large network into smaller networks called subnets. In this assignment, the network 192.168.10.0/24 is divided into four equal subnets using the /26 prefix. Each subnet has a different network address, broadcast address, and valid host IP range.

➢ **Subnet Plan Diagram**

- The following diagram shows the four subnets, their IP address ranges, and the devices connected to each subnet.

➢ Below Given Subnet Plan Diagram.




**Subnet Plan Summary**

| Subnet | Network Address | Subnet Mask | Broadcast Address | Valid Host Range |
|---|---|---|---|---|
| 1 | 192.168.10.0/26 | 255.255.255.192 | 192.168.10.63 | 192.168.10.1 – 192.168.10.62 |
| 2 | 192.168.10.64/26 | 255.255.255.192 | 192.168.10.127 | 192.168.10.65 – 192.168.10.126 |
| 3 | 192.168.10.128/26 | 255.255.255.192 | 192.168.10.191 | 192.168.10.129 – 192.168.10.190 |
| 4 | 192.168.10.192/26 | 255.255.255.192 | 192.168.10.255 | 192.168.10.193 – 192.168.10.254 |

**Device IP Assignment (Subnet 1 Example)**

| Device | IP Address |
|---|---|
| Router/Gateway | 192.168.10.1 |
| Laptop 1 | 192.168.10.2 |
| Laptop 2 | 192.168.10.3 |
| Printer | 192.168.10.4 |
| Mobile | 192.168.10.5 |

Legend: Router, Switch, Laptop, Printer, Mobile

**Notes:**
1. Each subnet uses /26 CIDR notation with subnet mask 255.255.255.192.
2. Network and broadcast addresses are not assigned to any device,
3. All devices in a subnet use the same subnet mask.

**(Subnet Plan Diagram for 192.168.10.0/24)**

## ➢ Result

- The network was divided into four equal subnets. Each subnet has 62 usable host addresses, and the devices are assigned IP addresses from their respective valid host ranges.

❖ **4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.**

➢ **Aim**

- To document the step-by-step process of calculating the subnet mask, determining the number of hosts per subnet, and assigning valid IP addresses to devices.

➢ **Introduction**

- Subnetting is the process of dividing a large network into smaller networks called subnets. It helps organize devices, manage IP addresses efficiently, and reduce unnecessary network traffic.
- In this assignment, the network 192.168.10.0/24 is divided into four equal subnets. The subnet mask, number of hosts, and IP addresses are calculated and assigned step by step.

➢ **Given Network Information**

- Original Network Address: 192.168.10.0
- Original CIDR Prefix: /24
- Original Subnet Mask: 255.255.255.0
- Required Number of Subnets: 4

➢ **Step 1: Calculate the Number of Borrowed Bits**

➢ First, we determine how many bits must be borrowed from the host portion to create four subnets.

➢ The formula is:

➢ Number of Subnets = (2^n)

➢ Here, (n) represents the number of borrowed bits.

- We need four subnets:
- (2^2 = 4)
- Therefore, we need to borrow 2 bits from the host portion.

- **Step 2: Calculate the New CIDR Prefix**
- The original CIDR prefix is /24.
- We add the two borrowed bits:
- New CIDR = 24 + 2 = /26
- Therefore, the new subnet prefix is /26.

- **Step 3: Calculate the New Subnet Mask**
- The original subnet mask is:
- 255.255.255.0
- After borrowing two bits, the new subnet mask becomes:
- 255.255.255.192
- The subnet mask in CIDR notation is /26.

- **Step 4: Calculate the Number of Hosts per Subnet**
- An IPv4 address contains 32 bits.
- Host bits = 32 − 26 = 6
- The total number of IP addresses in each subnet is:
- (2^6 = 64)
- Two addresses are reserved: one for the network address and one for the broadcast address.
- Usable Hosts = 64 − 2 = 62
- Therefore, each subnet contains 64 total IP addresses and 62 usable host addresses.

- **Step 5: Calculate the Subnet Block Size**
- The last octet of the subnet mask is 192.
- Block Size = 256 − 192
- Block Size = 64
- Therefore, the network address increases by 64 each time.
- The four subnet network addresses are:
  1. 192.168.10.0
  2. 192.168.10.64
  3. 192.168.10.128
  4. 192.168.10.192

- **Step 6: Identify Network and Broadcast Addresses**

| Subnet | Network Address | Broadcast Address |
|---|---|---|
| 1 | 192.168.10.0 | 192.168.10.63 |
| 2 | 192.168.10.64 | 192.168.10.127 |
| 3 | 192.168.10.128 | 192.168.10.191 |
| 4 | 192.168.10.192 | 192.168.10.255 |

- The network address identifies each subnet, while the broadcast address is used to send data to all devices within that subnet. Neither address can be assigned to an ordinary host device.

- **Step 7: Calculate the Valid Host Ranges**

| Subnet | First Usable IP | Last Usable IP |
|---|---|---|
| 1 | 192.168.10.1 | 192.168.10.62 |
| 2 | 192.168.10.65 | 192.168.10.126 |
| 3 | 192.168.10.129 | 192.168.10.190 |
| 4 | 192.168.10.193 | 192.168.10.254 |

- Each subnet has 62 usable host addresses.

- **Step 8: Assign IP Addresses to Devices**
- For this example, we assign five devices to the first subnet, 192.168.10.0/26.

| Device | Assigned IP Address | Subnet Mask |
|---|---|---|
| Router/Gateway | 192.168.10.1 | 255.255.255.192 |
| Laptop 1 | 192.168.10.2 | 255.255.255.192 |
| Laptop 2 | 192.168.10.3 | 255.255.255.192 |
| Printer | 192.168.10.4 | 255.255.255.192 |
| Mobile Phone | 192.168.10.5 | 255.255.255.192 |

- All five IP addresses are unique and belong to the valid host range of the first subnet.

- **Step 9: Verify the IP Address Assignments**
- Before finalizing the subnet plan, we check the following points:
  1. All devices have unique IP addresses.
  2. All assigned addresses belong to the correct subnet.
  3. The network address 192.168.10.0 is not assigned to a device.
  4. The broadcast address 192.168.10.63 is not assigned to a device.
  5. All devices use the subnet mask 255.255.255.192.
- These checks help prevent IP address conflicts and incorrect network configuration.

- **Result**
- The network 192.168.10.0/24 was successfully divided into four equal subnets using the /26 prefix. The subnet mask, total and usable host addresses, network addresses, broadcast addresses, and valid host ranges were calculated. Five devices were assigned unique and valid IP addresses in the first subnet.